

DDTA BA4-T2 method-owned interpretation, coverage loss and cross-projection consistency pressure test - R1

Status: COMPLETED / PROVISIONAL PASS WITH INTERPRETATION-COVERAGE REFINEMENT

Executed against repository baseline: f90ef3a0bc0b7712cb8081165c28e8923aec9e2d

Input candidate: BA4_PROJECTION_BOUNDARY_TRACEABILITY_CANDIDATE_R1.md

1. Question under test

Can the BA4-T1 projection boundary survive two intentionally incompatible method-oriented projections, aggressive information loss, non-current BA state and cross-baseline rebuild without either importing consumer semantics into Base Analysis or inventing a shared projection ontology?

The test is falsification-first. Each probe tries to force a semantic responsibility into the shared core. A responsibility is accepted only when removing it creates a concrete loss of reviewability, coverage honesty, qualification or traceability.

2. Fixed dependencies and guardrails

Fixed entering T2:

```
BA0    CLOSED
BA1    CLOSED
BA2    CLOSED
BA3    CLOSED
BA4-T1 provisional projection lower bound active
```

T2 does not define:

- a complete STRIDE or STRIDE-AI schema;
- a DFD ontology;
- AnalysisRecord or Common Finding;
- ThreatForge classes/tables/UI;
- a universal projection taxonomy;
- BA5 lexical rules; or
- BA6 final holdout closure.

Method labels used below are bounded pressure instruments only.

3. Test basis

The main facial-access BA basis includes the already-reviewed meanings:

```
transfer(CameraSubsystem,
         RecognitionProcessor,
         RecognitionCapture)

correlate(RecognitionCapture, RecognitionRequest)

consumeService(project, LocalConnectivity)

assignResponsibility[negative](
  project,
  underlyingTransport,
```

ownership/management)

realize(LocalConnectivity, WiredEthernet)

constrain(delivery, completion/security properties)

Controls use facial mutations M1-M4 plus order/WMS/provider meanings, including project-owned versus external inventory responsibility and governed `PaymentAuthorizationResult` normalization.

4. Probe P1 - incompatible method taxonomies

Hypothesis attacked

Two method projections need a shared category ontology to remain consistent.

Projection F

Local kinds:

`FlowParticipant`

`FlowExchange`

`ExposureAnnotation`

It selects transfer/service-use meaning and may interpret service-use plus negative responsibility as:

`externally-provided-transport-exposure`

Projection A

Local kinds:

`AssuranceSubject`

`AuthorityPlacement`

`ContractDependency`

`NormalizationExpectation`

It selects responsibility/contract/normalization meaning and may interpret the same service/responsibility basis as:

`delegated-assurance-dependency`

Result

PASS without shared taxonomy.

The two items can be compared through overlapping role-bound BA trace. Their category difference is method-local, not a contradiction.

Disposition

universal shared projection ontology	REJECTED
BA trace as common semantic anchor	PASS

5. Probe P2 - descriptor plus trace reproducibility

Hypothesis attacked

An immutable descriptor plus role-bound BA trace is always sufficient to reconstruct method-owned interpretation.

Counterexample

One immutable projection revision contains two local rules whose inputs overlap:

```
R-ext-service
  serviceUse + responsibilityBoundary
    -> externally-provided-transport-exposure

R-assurance-review
  serviceUse + responsibilityBoundary + constraintSet
    -> assurance-dependency-review
```

An item with only `serviceUse` and `responsibilityBoundary` trace identifies its source basis but not which local rule created its method-owned meaning when several rule paths remain valid under the same descriptor.

Result

The T1 lower bound is insufficient for non-trivial method interpretation.

Smallest refinement

```
methodOwnedInterpretation present
  -> interpretationRuleRef required
```

The referenced local rule revision is already contained/resolved under the immutable projection descriptor revision.

Negative control

No BA3 derivation rule is reused. The interpretation is not method-neutral and cannot create accepted BA meaning.

6. Probe P3 - aggressive lossy selection**Hypothesis attacked**

A generic coverage statement can represent both intentionally selective and complete-within-scope projections.

Counterexample

Two views have the same eligible BA scope:

```
current accepted transfer propositions
```

but one promises to render all eligible transfers while the other intentionally chooses only transfers touching one analysis focus.

If both use only the statement `transfer propositions are in scope`, an omitted transfer is ambiguous:

- incomplete projection build; or
- valid selective omission.

Result

T1 coverage contract requires an explicit mode:

EXHAUSTIVE_FOR_DECLARED_SCOPE
SELECTIVE

Important boundary

An omission from an exhaustive projection is a projection completeness defect. It is not a project-level negative proposition and does not prove source documentation incompleteness.

7. Probe P4 - omission as negative fact

Hypothesis attacked

If an exhaustive projection omits an item, the consumer may treat the missing meaning as false in the project.

Result

REJECTED.

Even under exhaustive coverage, the claim is only:

this materialization failed its declared projection contract

The source of project truth remains governed documentation through accepted BA.

8. Probe P5 - diagnostic/stale review projection

Hypothesis attacked

A projection can expose stale or diagnostic BA meaning using the same rendering semantics as current accepted BA.

M4 pressure

Representation context changes while an older transfer source remains inconsistent. BA3 can mark the old transfer STALE and localize a DIAGNOSTIC_UNRESOLVED condition.

Two projection purposes are tested:

normal current-state method view

qualificationPolicy = ACCEPTED + CURRENT

review/change view

qualificationPolicy includes STALE / DIAGNOSTIC_UNRESOLVED

Result

The review projection is valid only when the non-current/unresolved qualification remains visible in the consumed meaning.

The diagnostic is not promoted to project fact merely because it is accepted as a diagnostic.

Smallest refinement

qualificationPolicy is explicit inside the projection coverage contract.

No duplicate BA3 lifecycle state machine is introduced.

9. Probe P6 - interpretation over stale basis

Hypothesis attacked

A method-owned annotation may be presented as a current result even when one of its BA inputs is stale.

Result

REJECTED for current shared interpretation.

A review-oriented method projection may still compute the local annotation to identify potentially affected analysis, but it must preserve that its basis is stale/unresolved.

No new universal projection status enum is forced; the projection descriptor and trace to BA3-qualified inputs are sufficient.

10. Probe P7 - cross-projection consistency

Hypothesis attacked

Two projections over the same BA must have equivalent categories/items to be considered consistent.

Result

REJECTED.

Consistency is checked against BA, not by pairwise method taxonomy equality.

For the same BA baseline:

Projection F item

```
basis = {consumeService, negative responsibility}
interpretation = externally-provided-transport-exposure
```

Projection A item

```
basis = {consumeService, negative responsibility}
interpretation = delegated-assurance-dependency
```

Both are valid if each is allowed by its own immutable local interpretation rule and remains explicitly method-owned.

Comparison rule

cross-projection common denominator = BA trace

No first-class method-to-method equivalence identity is required.

11. Probe P8 - contradictory shared rendering

Hypothesis attacked

Different projections can disagree about shared BA truth as long as their methods differ.

Counterexample

BA contains negative project responsibility for underlying transport.

Projection F renders:

```
project does not own/manage underlying transport
```

Projection A renders as shared truth:

`project owns underlying transport`

Result

Projection A fails semantic preservation. Method diversity does not excuse inversion of shared BA polarity.

No cross-projection contradiction proposition is needed; the failing projection can be diagnosed directly against its traced BA basis.

12. M1 replay - realization change

B0:

`realize(LocalConnectivity, WiredEthernet)`

B1:

`realize(LocalConnectivity, WiFi)`

Projection F ignores realization technology under its declared scope and can rebuild with the same flow meaning.

Projection A includes realization/assurance context and changes local output.

PASS: different projection deltas are expected because coverage differs.

13. M2 replay - responsibility change

When transport becomes project-owned:

- negative responsibility is replaced by affirmative responsibility;
- external service consumption may retire;
- Projection F's external-service exposure rule ceases to apply;
- Projection A can emit a different authority-placement item.

PASS: local categories evolve through rebuild from BA, not through BA mutation from the projections.

14. M3 replay - remote to local recognition

The remote transfer is retired by BA.

An exhaustive current-flow projection must omit the old exchange in B1 because it is no longer eligible current BA meaning.

A method view retaining the B0 flow merely because its local item existed fails rebuild semantics.

PASS: no cross-baseline projection item identity is required.

15. M4 replay - stale/unresolved representation

A current projection excludes stale capture transfer under its qualification policy.

A review projection may include:

`stale transfer`

`unresolved representation conflict`

with explicit qualification.

REJECTED controls:

- silently rewrite transfer payload from sibling Decision;
- show stale transfer as current;
- use diagnostic as new governed project fact.

16. Order/WMS replay

Externalizing inventory authority toward a WMS produces different local effects:

flow/integration projection
new service/interchange focus

assurance/responsibility projection
moved authority/contract dependency focus

Unrelated payment/fulfillment BA meaning is not invalidated merely because one projection changes strongly.

PASS: method deltas remain localized by BA trace and coverage.

17. Provider-normalization replay

A method that needs `PaymentAuthorizationResult` must consume the accepted BA normalized result.

The method may then derive a local branch or assurance category under its own interpretation rule.

REJECTED:

raw provider state
-> private method mapping
-> projected value presented as shared governed result

PASS: BA3 remains the authority-preserving normalization layer.

18. Cross-baseline and cross-projection comparison

T2 tries to force a new shared identity for projection items across methods and baselines.

The requirement does not survive.

Same-baseline comparison uses role-bound BA trace. Cross-baseline comparison first resolves BA3 continuity, then rebuilds each projection against the target BA baseline.

Projection-local continuity may still be useful to a UI/tool, but current DDTA evidence does not require it as a Base Analysis projection semantic responsibility.

19. Attempts to force earlier reopen

BA1

No method-local node/category requires new shared project-semantic identity with subtype-specific invariants.

BA1 reopen = NOT TRIGGERED.

BA2

No method-owned category exposes missing methodology-neutral project meaning requiring a new operator/role.

BA2 reopen = NOT TRIGGERED.

BA3

Source provenance, review/freshness and BA continuity remain sufficient to qualify projection input and rebuild after change.

BA3 reopen = NOT TRIGGERED.

20. T2 dispositions

Two incompatible method taxonomies	PASS
Shared projection ontology	REJECTED
Method interpretation promoted into BA	REJECTED
Descriptor + trace alone for multiple local rules	REJECTED
Interpretation rule reference	REQUIRED
Role-bound BA trace	RETAINED
Selective/exhaustive coverage collapse	REJECTED
Coverage mode	REQUIRED
Exhaustive omission -> project negation	REJECTED
Exhaustive omission -> projection defect	REQUIRED
Qualification policy	REQUIRED
Diagnostic/stale review projection	PASS
Non-current BA presented as current shared truth	REJECTED
Cross-projection taxonomy equality	NOT REQUIRED
Cross-projection comparison through BA trace	PASS
New shared projection identity	NOT FORCED
Projection item BA-like lifecycle	NOT FORCED
Universal projection DSL	REJECTED
New BAE family	NOT FORCED
New BA2 operator	NOT FORCED
BA1 / BA2 / BA3 reopen	NOT TRIGGERED
BA4-T1	NOT REOPENED
BA4	STARTED / NOT CLOSED

21. Provisional conclusion

BA4-T2 does not falsify the T1 projection boundary, but it does falsify two under-specified parts of the lower bound:

1. method-owned interpretation needs item-to-rule accountability when several local rules can use overlapping BA bases; and
2. coverage needs to distinguish selective views from views that promise exhaustive representation within an inspectable declared BA scope.

The accepted refinement remains projection-local and does not expand BA ontology.

22. Smallest remaining BA4 question

The next step should be an integrated closure review, not another method taxonomy exercise.

It must try to remove the T2 refinements, regress the combined T1/T2 contract across both corpora and determine whether BA4 can close for current thesis scope.

23. Next authorized microstep

Only after this T2 package is reviewed, committed, pushed and remotely verified:

BA4-T3 - projection boundary, interpretation/coverage and cross-projection closure review.

Do not begin BA5 before BA4 is explicitly closed.